IoDT2 Thing Description - User Guide

This guide walks you through creating, managing, and searching digital twin (Thing) definitions using the IoDT2 Thing Description platform's web interface.

Access the application at: http://localhost:3005 **API documentation:** http://localhost:3015/docs

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Table of Contents

1. Interface Overview
2. Tenant Management
3. Creating a Thing
4. Thing List
5. Searching Things
6. Thing Details Page
7. Exporting
8. Language and Theme Settings
9. Quick Start Scenario
10. Quick Reference Table

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1. Interface Overview

The application consists of three main sections:

```
+-------------------------------------------------------+
|  Navbar (Top Bar)                                     |
|  [Logo]  [Tenant Selector]       [Language] [Theme]   |
+----------+--------------------------------------------+
|          |                                            |
| Sidebar  |  Main Content Area                         |
|          |                                            |
| - List   |  (Selected page content is displayed here) |
| - Create |                                            |
| - Search |                                            |
|          |                                            |
+----------+--------------------------------------------+
```

Navbar (Top Bar)

- **Logo** — Displays "IoDT2 Demo" and navigates to the home page.
- **Tenant Selector** — Shows the currently active tenant and allows you to switch between tenants.
- **Language Selector** — Switches the interface between Turkish and English.
- **Theme Button** — Toggles between Light and Dark display modes.

Sidebar (Left Menu)

Three main navigation items are available:

Menu Item	Page	Description
Thing List	`/things`	Lists all created Things
Create New Thing	`/things/create`	Opens the Thing creation form
Search	`/things/search`	Advanced search page

The sidebar is collapsible — use the arrow button at the bottom left to collapse or expand it.

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2. Tenant Management

The system uses a multi-tenant architecture. Each tenant has its own Thing collection, and data is kept isolated between tenants.

Selecting a Tenant

1. Click the **Tenant Selector** dropdown in the Navbar.
2. Choose the tenant you want to work with from the list.
3. The selected tenant becomes active; all operations (listing, creating, searching) are scoped to this tenant.

   Note: The default tenant is `default`. Thing IDs are automatically prefixed with the tenant name (e.g., `default:sensor-01`).

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3. Creating a Thing

Navigation: Sidebar > **Create New Thing**

3.1 Selecting a Thing Type

The first step is to select the Thing type. Each type serves a different purpose:

Type	Description	Example
Sensor	Represents a single independent sensor. Suitable for data-collecting devices.	Temperature sensor, humidity sensor
Device	Contains multiple sensors or actuators within a single unit.	Weather station (temperature + humidity + pressure)
Component	Part of a hierarchical structure, connected to other Things via relationships.	A floor of a building, a gateway's sub-component

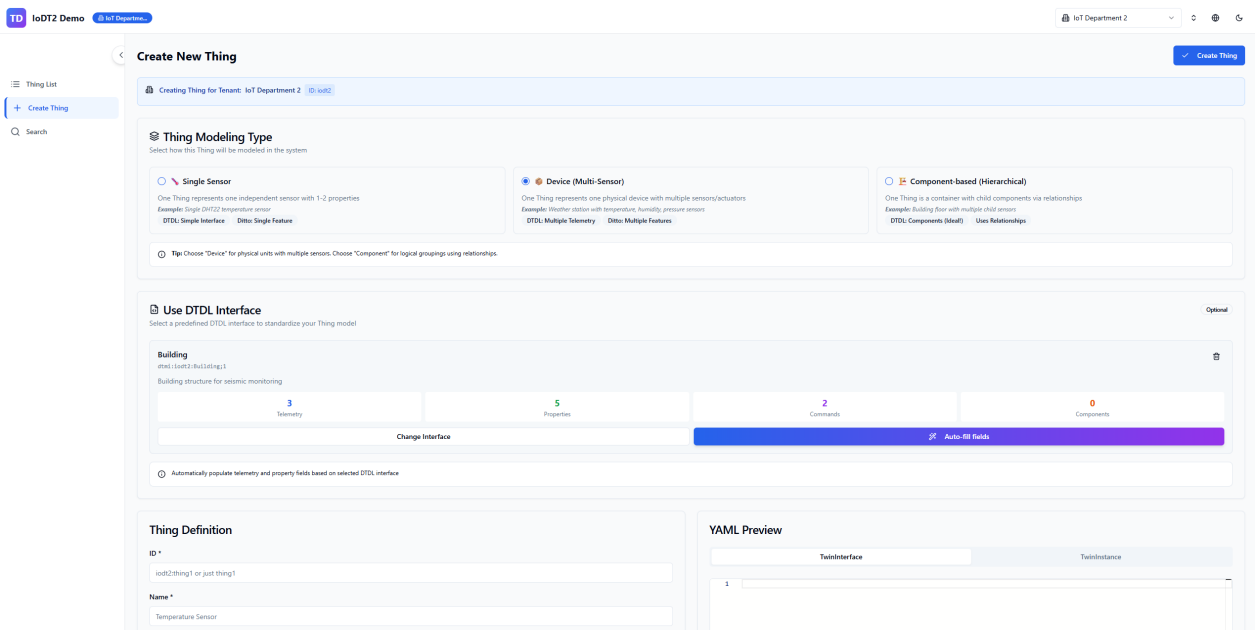

Figure 1: Create Thing

3.2 DTDL Interface Binding (Optional)

DTDL (Digital Twins Definition Language) is a standard schema definition language for digital twins. Binding a DTDL interface automatically populates your Thing's properties.

Steps:

1. Click the **Select DTDL Interface** button.
2. In the modal that opens, filter interfaces by category or keyword:
   - Categories: Environmental, Seismic, Air Quality, etc.
   - Use the search bar to filter by name
3. Select an interface — a summary appears on the right:
   - Telemetry count
   - Property count
   - Command count
   - Component count
4. Click **Select** to confirm.
5. The form is automatically filled with properties from the DTDL interface.

   You can later **Change** the selected DTDL interface or **Remove** it entirely.

3.3 Defining Contents (Tabs)

The form in the left panel is organized into four tabs:

Properties Tab Fill in the following fields for each property:

Field	Description	Example
Name	Property name	`temperature`
Type	Data type: String, Integer, Float, Boolean, Object, Array	`Float`
Description	Description text (optional)	`Ambient temperature`
Writable	Is it writable? (checkbox)	Enabled/Disabled
Minimum	Lower bound value (optional)	`-40`
Maximum	Upper bound value (optional)	`85`
Unit	Unit of measurement (optional)	`celsius`

- Click **Add Property** to add a new property.
- Click the trash icon on a row to delete it.
- Properties inherited from the DTDL interface appear automatically.

Relationships Tab Define connections between Things:

Field	Description	Example
Name	Relationship name	`installedIn`
Target Interface	Target Thing's interface name	`Building`
Description	Relationship description (optional)	`Building where the sensor is installed`

Commands Tab Define commands that the Thing can execute:

Field	Description	Example
Name	Command name	`reboot`
Description	Command description	`Restart the device`

Commands can be edited in two modes: - **Simple Mode** — Using form fields - **Advanced Mode** — Schema definition via JSON editor

Location Tab Set the Thing's physical location on an interactive map:

1. Click on the desired point on the map.
2. **Latitude** and **Longitude** are filled in automatically.
3. The **Address** field is populated automatically via reverse geocoding.
4. The **Altitude** value is calculated automatically.

   You can also enter coordinates manually.

3.4 Domain Metadata (Device Information)

Expand the metadata section to enter the following information:

Field	Description
Manufacturer	Manufacturer name
Model	Device model
Serial Number	Serial number
Firmware Version	Firmware version

3.5 YAML Preview and Validation

The right panel contains two tabs that update in real time:

- **Interface YAML** — Shows the Thing's interface definition (type, properties, relationships, commands).
- **Instance YAML** — Shows the Thing instance definition (linked interface, metadata).

The YAML preview is rendered in a Monaco Editor with syntax highlighting.

3.6 DTDL Validation Panel

If a DTDL interface is selected, a validation panel appears at the bottom of the right panel:

- **Compatibility Score** — Shows how compatible the Thing is with the DTDL interface as a percentage.
- **Matched Properties** — Lists which DTDL properties/telemetry have counterparts in the Thing.
- **Missing Properties** — Shows properties defined in DTDL that are not present in the Thing.
- **Validation Issues** — Lists type mismatches or other warnings.

3.7 Saving

After entering all information, click the **Create** button at the top of the page. Upon successful creation, the Thing is saved to the Fuseki RDF store and you are redirected to the Thing List page.

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4. Thing List

Navigation: Sidebar > **Thing List**

All Things are displayed in a card view. Each card shows:

- Thing name
- Description text
- Thing type badge (Sensor / Device / Component)
- Creation date

Actions

Action	Icon/Button	Description
Search	Search bar at the top	Filters Things by name
Create New	**+** button	Redirects to the creation page

Action	Icon/Button	Description
View	Eye icon	Opens the Thing details page
Export	Download icon	Downloads as a ZIP file
Delete	Trash icon	Permanently deletes after confirmation

Empty State

If no Things have been created yet, the page displays an informational message with a link to the creation form.

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5. Searching Things

Navigation: Sidebar > **Search**

The search page operates in three different modes. Switch between modes using tabs.

5.1 Standard Search

A free-text search. Type a keyword into the search bar.

Open the **Advanced Filters** section to narrow your search:

Filter	Description	Example
Domain	Domain category	Environmental, Seismic, Air Quality
Type	Thing type	Sensor, Device, Component
Status	Operational status	Active, Inactive
Location	Geographic area	Istanbul, Ankara
Properties	Existence of a specific property	`temperature`, `humidity`

5.2 Value Search

Allows you to search based on a specific property's value.

Steps:

1. Select a property from the **Property** dropdown (e.g., `temperature`).
2. Choose a comparison operator from the **Operator** dropdown:
   - = (equals)
   - > (greater than)
   - < (less than)
   - >= (greater than or equal)
   - <= (less than or equal)
   - != (not equal)
3. Enter the value in the **Value** field (e.g., 25).
4. Click the **Search** button.

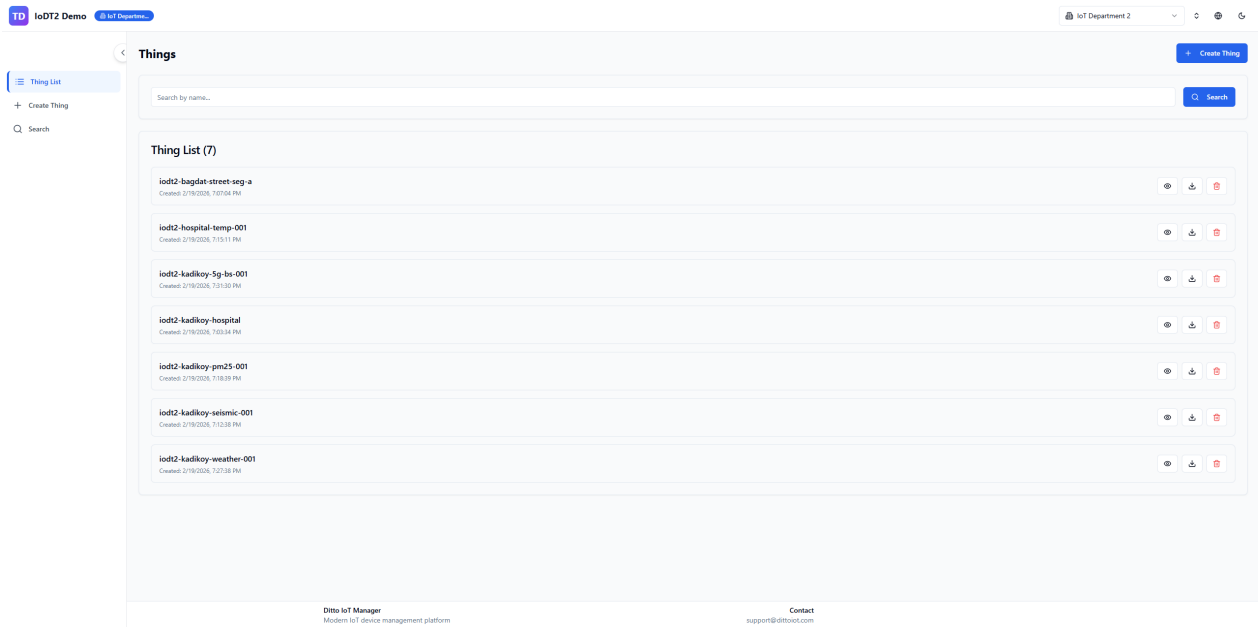

Figure 2: Thing List

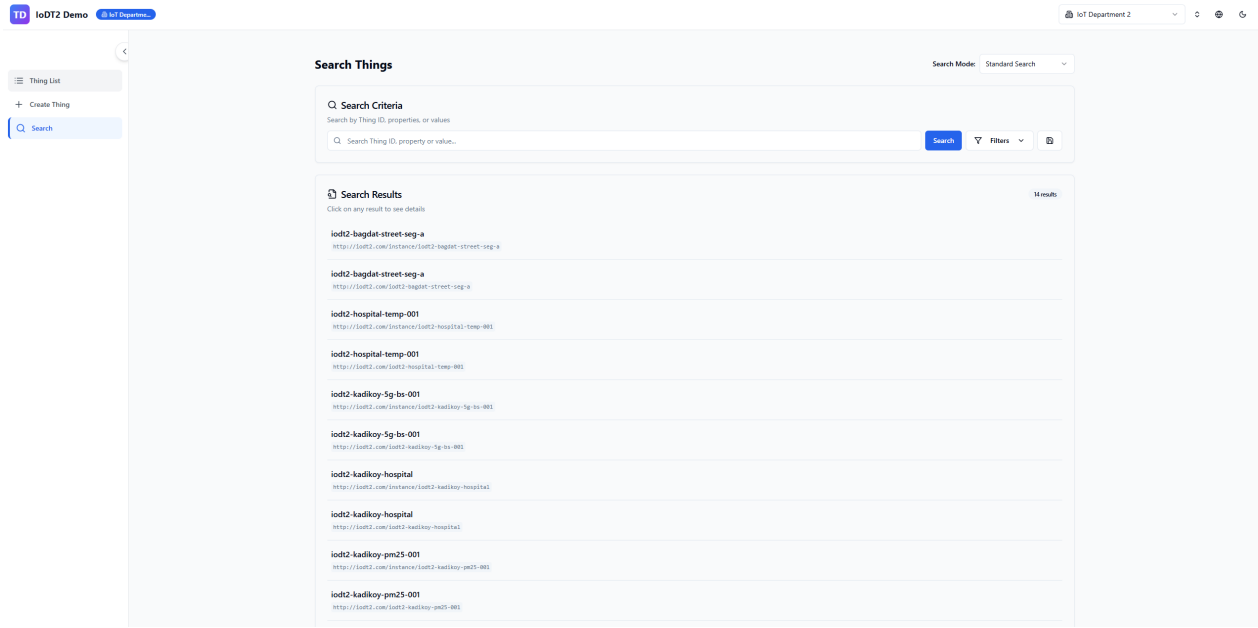

Figure 3: Search Things

Example: `temperature > 25` — lists all sensors whose temperature upper bound is greater than 25.

5.3 SPARQL Search

Allows you to write SPARQL queries directly against the RDF database. Designed for advanced users.

Interface Features:

- **Monaco Editor** — Write SPARQL queries with syntax highlighting.
- **Templates** button — Lists ready-made query templates. Click a template to insert it into the editor.
- **Run** button — Executes the query.
- **Copy** button — Copies the query to the clipboard.
- **Export** button — Downloads the query as a file.

Available Prefixes:

The following prefixes are automatically injected into queries:

```
PREFIX ts:   <http://twin.dtd/ontology#>
PREFIX tsd:  <http://iodt2.com/>
PREFIX rdf:  <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd:  <http://www.w3.org/2001/XMLSchema#>
```

Example SPARQL Queries:

List all Thing interfaces:

```
SELECT ?interface ?name ?description
WHERE {
  GRAPH ?g {
    ?interface a ts:TwinInterface .
    ?interface ts:name ?name .
    OPTIONAL { ?interface ts:description ?description }
  }
}
ORDER BY ?name
```

Find Things with a specific property:

```
SELECT ?name ?propName ?propType ?unit
WHERE {
  GRAPH ?g {
    ?thing a ts:TwinInterface .
    ?thing ts:name ?name .
    ?thing ts:hasProperty ?prop .
    ?prop ts:propertyName ?propName .
    ?prop ts:propertyType ?propType .
    OPTIONAL { ?prop ts:unit ?unit }
```

```
  }
  FILTER(?propName = "temperature")
}
```

List writable properties:

```
SELECT ?thingName ?propName ?unit
WHERE {
  GRAPH ?g {
    ?thing a ts:TwinInterface .
    ?thing ts:name ?thingName .
    ?thing ts:hasProperty ?prop .
    ?prop ts:propertyName ?propName .
    ?prop ts:writable true .
    OPTIONAL { ?prop ts:unit ?unit }
  }
}
```

5.4 Search History and Saved Searches

- **Search History** — Your searches are saved automatically. Click on a previous search to re-execute it.
- **Save Search** — Save a search with a custom name. Saved searches appear in the list and can be re-executed with a single click.

5.5 Results

Search results are displayed in the same card format as the Thing List. Each result card supports viewing, exporting, and deleting.

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6. Thing Details Page

Access: Click on any Thing from the Thing List or search results.

The details page consists of four tabs:

Properties Tab

All defined properties are displayed in a table:

Column	Description
Name	Property name
Schema Type	Data type (string, float, integer, etc.)
Value	Current value (if assigned)
Description	Description

Relationships Tab

Shows connections to other Things:

Column	Description
Name	Relationship name
Target Interface	Target Thing interface

Commands Tab

Lists executable commands:

Column	Description
Name	Command name
Description	Command description

DTDL Binding Tab

If the Thing is bound to a DTDL interface, detailed information is displayed in this tab:

- **Bound Interface Info** — Interface name, DTMI (Digital Twin Model Identifier), category, thing-type.
- **Property Mapping** — For each DTDL property/telemetry:
  - Matched Thing property
  - Unit
  - Writable status
- Information is presented visually as badges.

Page Actions

Action	Description
Export	Downloads the Thing definition as a ZIP file
Delete	Permanently deletes the Thing (requires confirmation)
Back	Returns to the previous page

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7. Exporting

When you click the **Export** (download icon) button from the Thing List or Details page, the Thing definition is downloaded as a ZIP file.

ZIP file contents:

File	Description
`interface.yaml`	Thing interface definition (properties, relationships, commands)
`instance.yaml`	Thing instance definition (metadata, binding information)

These files contain the complete Thing definition and can be used for transferring to other systems or for backup purposes.

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8. Language and Theme Settings

Changing Language

Use the **language dropdown** on the right side of the Navbar to change the interface language: - **English** — English interface - **Turkce** — Turkish interface

All menu, button, and label texts are updated according to the selected language.

Changing Theme

Use the **theme button** in the Navbar to change the display mode: - **Light Mode** — Light background display - **Dark Mode** — Dark background display (reduces eye strain)

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9. Quick Start Scenario

The following steps demonstrate the process of creating a temperature sensor from scratch.

Scenario: Creating a Temperature Sensor

1. Select a tenant - Select the `default` tenant from the Tenant Selector in the Navbar.

2. Navigate to the creation page - Click **Create New Thing** from the Sidebar.

3. Enter basic information - Thing Name: `Office Temperature Sensor 01` - Thing Type: Select **Sensor**.

4. Bind a DTDL interface - Click **Select DTDL Interface**. - Select **Environmental** from the categories. - Choose the **TemperatureSensor** interface and confirm with **Select**. - Properties (temperature, humidity, etc.) will be automatically populated in the form.

5. Set the location - Switch to the **Location** tab. - Click on the map at the point where the sensor is installed. - Address and coordinates are filled in automatically.

6. Enter device information (optional) - Open the Metadata section. - Manufacturer: `Bosch` - Model: `BME280`

7. Review the YAML preview - Check the Interface and Instance YAML tabs in the right panel. - Verify that the compatibility score is high in the DTDL validation panel.

8. Save - Click the **Create** button. - After the success message, you can see your new sensor in the Thing List.

Next Steps

- Go to the **Thing List** to view your sensor.
- Query your sensor using SPARQL from the **Search** page:

```
SELECT ?name ?propName ?unit
WHERE {
  GRAPH ?g {
    ?thing a ts:TwinInterface .
    ?thing ts:name ?name .
    ?thing ts:hasProperty ?prop .
    ?prop ts:propertyName ?propName .
    OPTIONAL { ?prop ts:unit ?unit }
  }
  FILTER(?name = "Office Temperature Sensor 01")
}
```

- **Export** the YAML files to download and review them.

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10. Quick Reference Table

Task	Navigation
Create a Thing	Sidebar > Create New Thing
View all Things	Sidebar > Thing List
Search Things	Sidebar > Search
View Thing details	Thing List > Eye icon on a card
Export Thing (ZIP)	Thing List or Details > Download icon
Delete a Thing	Thing List or Details > Trash icon
Run SPARQL query	Search > SPARQL tab
Change tenant	Navbar > Tenant Selector
Change language	Navbar > Language dropdown
Change theme	Navbar > Theme button
API documentation	http://localhost:3015/docs